

6. Humans are protected against disease by vaccination programmes and the use of antibiotics.

- (a) Measles is a highly infectious viral illness that is uncommon in the UK because of the effectiveness of vaccination. The measles vaccine provides life-long protection against the disease.

(i) Explain how vaccines work.

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- (ii) A flu vaccine is also available. Unlike the measles vaccine, it needs to be given every year and is not 100% effective. Explain why a flu vaccine does not provide life-long protection.

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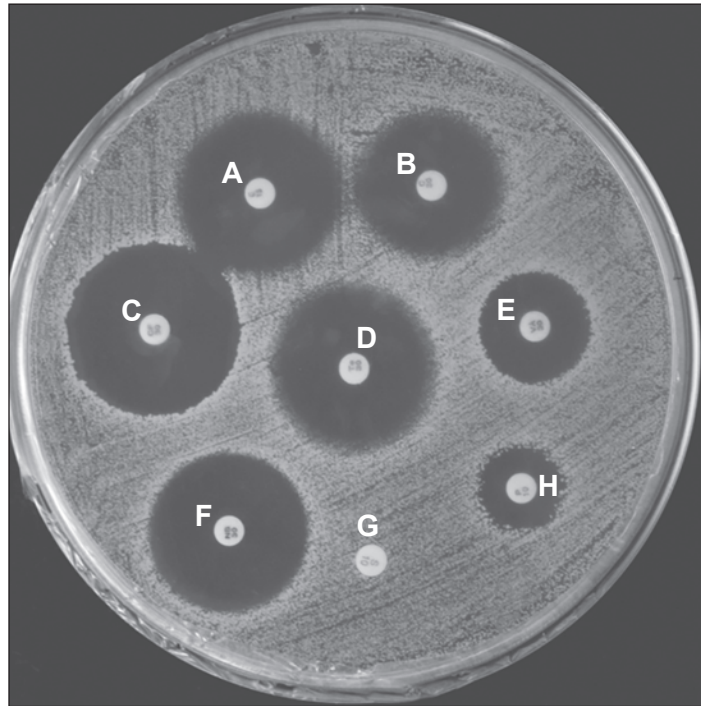
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- (b) Doctors prescribe patients antibiotics to help fight bacterial infections. There are a variety of antibiotics available.

The photograph below shows the results of an experiment to determine the effectiveness of different antibiotics against one type of bacterium.

Eight different antibiotics were tested. Each disc contains one of the antibiotics (**A** to **H**).



- (i) Evaluate the effectiveness of the antibiotics **A** to **H**.

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- (ii) A disc containing sterile water could be included to act as a control (not shown in the photograph). State the purpose of this control disc.

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